

Building **Evacuation** Navigation System

Complete Workflow & System Understanding Strategy

1. Input Stage: Building Data

Data Acquisition

- **Blueprints:** Digital architectural floor plans (CAD/PDF) are uploaded by Building Admins.
- **Management Portal:** A dedicated dashboard where Facility Managers define exit points and safe zones.
- **Digitization:** Converting static images into scalable vector coordinates for internal processing.



2. Processing: Identification



Structural Nodes

System identifies Rooms, Corridors, and Doors as discrete nodes in a spatial database.



Exits & Safety

Automated tagging of emergency exits, fire extinguishers, and assembly points.



Dynamic Hazards

Real-time fire location inputs (either via manual reporting or future IoT sensors).

2. Processing: Graph-Based Logic

 Graph Nodes Illustration

Pathfinding Algorithms

- **Dijkstra/BFS:** Used to find the absolute shortest path from a user's location to the nearest exit.
- **Edge Weighting:** Danger zones (smoke/fire) are assigned infinite "weight," forcing the algorithm to reroute.
- **Relational Mapping:** Corridors act as edges connecting various room nodes.

3. Emergency: **Dynamic Rerouting**

Scenario: Corridor Blockage

If a primary corridor catches fire, the system instantly "disables" that edge in the graph.

- **Immediate Recalculation:** Routes are recomputed in milliseconds.
- **Safety First:** Avoids "Dead End" corridors where smoke might accumulate.
- **Audible Alerts:** Voice navigation warns users: *"Corridor B blocked. Turn back."*

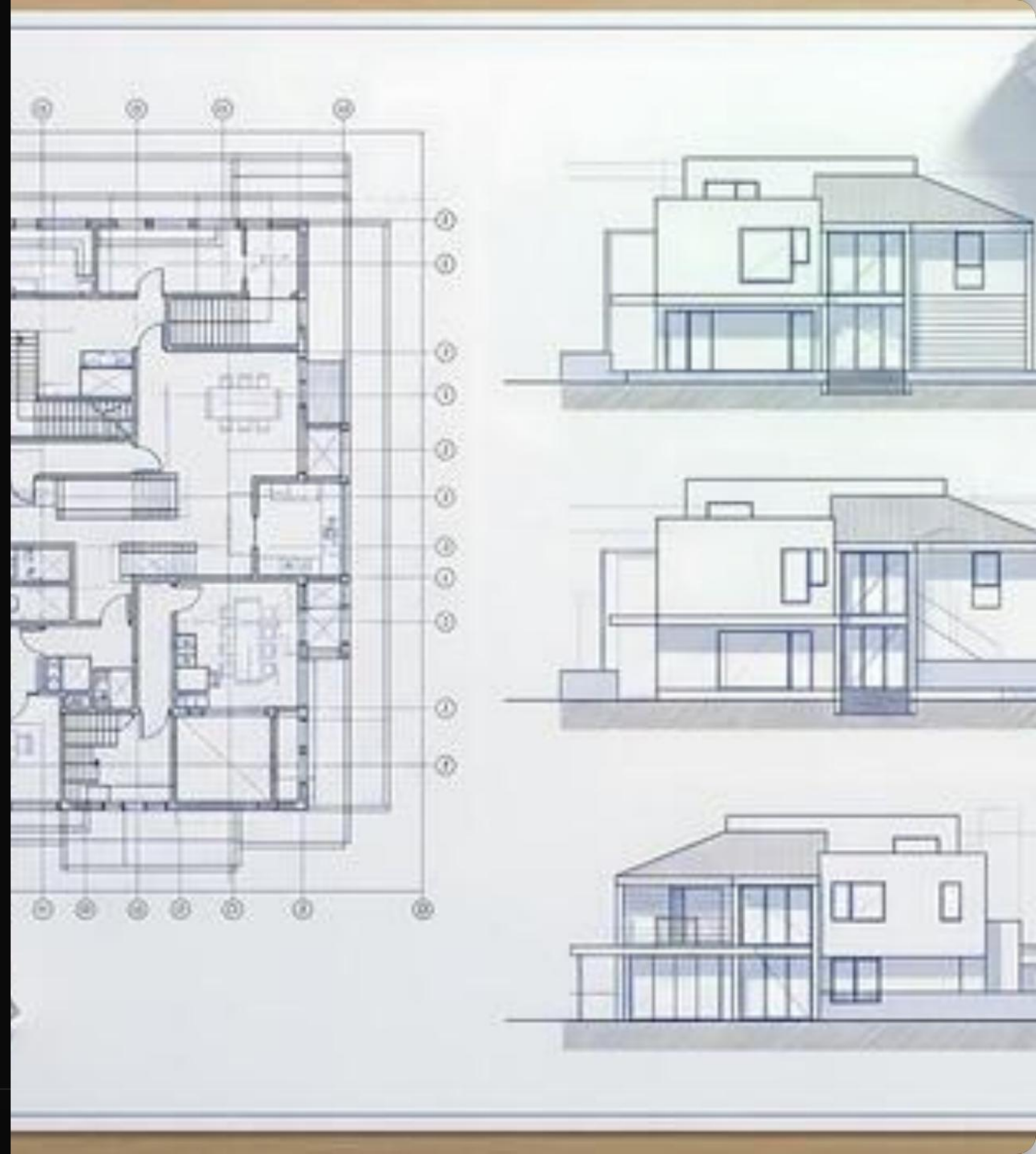
Emergency App UI

3. Emergency: **Multi-Floor Scaling**

Staircase Management

In multi-floor buildings, staircases are "Inter-Floor Bridges." If a staircase is unsafe, the system shifts traffic to the alternate stairwell.

Vertical and horizontal nodes are linked, ensuring a seamless flow from the 5th floor down to the ground exit.



4. Output: **User Experience**



Visual Path

A 2D high-contrast floor plan showing a bright green path through safe nodes.



Alert Banners

Red banners indicating exactly where the fire is located to ensure awareness.



Estimated Time

Displays time to reach the exit to keep users calm and informed.

5. Future Idea: IoT Integration



Live Sensor Data

- **Smoke/Heat Sensors:** Automated path rerouting without manual input.
- **Smart Lighting:** Physical floor lights could sync with the app's route to guide people in smoke.
- **Crowd Sensing:** Avoiding exits that are too crowded to prevent trampling.

5. Future Idea: QR & AI Mapping

Indoor Positioning

- **QR Tags:** Strategically placed QR codes to help the app "calibrate" user location instantly.
- **AI Blueprinting:** Using Computer Vision to automatically identify walls and doors from a photo of a blueprint.
- **Wearable Sync:** Sending heart rate and location to emergency responders for high-risk individuals.

 QR Positioning Concept

The Goal: Saving Lives

Combining Graph Theory with Real-Time Communication to eliminate panic and provide clarity in moments of crisis.